

Lecture 3: Broadcast Standards, Formats, Protocols and Data Types

In this lecture, we expand the broadcast media delivery pipeline by looking at specific protocols, formats, data types and technologies. The previous two lectures introduced the overall production, distribution and consumption stages, first conceptually and then through concrete processing blocks. This lecture maps actual industry standards and formats to those blocks, focusing on broadcast distribution.

The familiar three-stage model still applies: production, distribution and consumption. What changes now is the level of detail. Instead of only saying that content is packaged, multiplexed, transmitted, received and processed, we now identify the standards and data structures used at each step.

The emphasis is on broadcast TV, including terrestrial, satellite and cable broadcasting. We will also look at MPEG Transport Stream, PES packets, PSI/SI metadata tables, video and audio coding, subtitles, conditional access and receiver SoCs.

Broadcast Distribution Standards

Broadcast TV distribution differs depending on the physical transmission medium. Terrestrial, satellite and cable broadcasting each use different modulation and transmission technologies, but they commonly rely on MPEG-TS and related structures once the signal is recovered as a digital bitstream.

Terrestrial Broadcasting Standards

For terrestrial broadcasting, four major groups of standards are used around the world.

DVB-T, with its newer version DVB-T2, dominates in Europe, Africa and Australia, and is also used in several other countries such as India, Indonesia and Singapore.

ATSC standards are used primarily in the United States, Canada and South Korea.

ISDB-T is used in Japan and in many countries across South and Central America.

China uses its own terrestrial broadcasting standard called DTMB.

These standards focus mainly on the modulation techniques and configuration of terrestrial delivery. Each country defines its own frequency spectrum, frequency divisioning and configuration parameters. Receiver devices must therefore include tuners compliant with the standard used in the target country.

In other words, terrestrial broadcast standards are not only about the stream format. They also define how the signal is physically transmitted over the air and how receivers should lock to and demodulate it.

Container Formats in Broadcast TV

On the digital signal side, the first essential concept is the container format.

Digital streams are arrays of bits. A container format defines how those bits are organized into packets, how they are identified, how they are referenced and how elementary streams can be extracted for decoding and playback.

Container formats are not tied to a specific video or audio codec. They do not assume that the payload must be video, audio or any particular type of data. They provide a transport structure capable of carrying whatever data is needed.

In broadcast TV, MPEG-2 Transport Stream and PES are the de facto unified container formats.

PES stands for Packetized Elementary Stream. It defines how each elementary stream, such as video or audio, is packaged into PES packets. Each PES packet holds a decodable unit of data, depending on the stream type.

MPEG Transport Stream further splits PES packets into small fixed-size TS packets. Each TS packet is 188 bytes long. This small, constant packet size helps the broadcast stream maintain a steady timing structure, allowing transmission and reception to remain synchronized.

Metadata Formats: PSI, SI and PSIP

Broadcast streams also require metadata. Metadata tells the receiver what exists inside the multiplex and how to interpret it.

In MPEG broadcasting, the core metadata format is MPEG Program Specific Information, or PSI. DVB adds Service Information, or SI. ATSC defines its own related standard called PSIP, which stands for Program and System Information Protocol.

These standards define signalling tables. The tables are split into sections and packed into MPEG-TS packets, just like other data.

Each table has a specific name and purpose. Together, they tell the receiver which elementary components exist inside a multiplex, what their PIDs are, which coding formats are used, what the service names are, whether some streams are scrambled and other important information.

Without these tables, the receiver would see only a stream of TS packets. It would not know which packets belong to a specific service, which packets carry video, which carry audio, which carry subtitles, or which streams need descrambling.

Video Coding Formats

Video coding formats are among the most important standards in digital TV. However, from the host software perspective, their internal compression mechanisms are often abstracted away by hardware video decoders. Host software usually configures decoders and manages the flow, but it does not normally implement video decompression itself.

Broadcast video coding has historically been dominated by MPEG-related standards.

It started with MPEG-2 video. Later, H.264 Advanced Video Coding, also known as AVC, became widely used. H.265 High Efficiency Video Coding, or HEVC, further improved compression efficiency. The newer H.266 Versatile Video Coding, or VVC, continues this evolution.

These codecs define additional internal data structures used to represent compressed video. Examples include groups of pictures, frames, macroblocks, motion vectors and residuals.

A group of pictures can be packed into a PES packet and then transported through MPEG-TS. The receiver eventually extracts the video PES packets and sends them to the video decoder.

Audio Coding Formats

Audio streams also use dedicated compression and coding standards.

Common examples include AAC, or Advanced Audio Coding, HE-AAC, or High Efficiency AAC, and AC-3. Modern TV systems also support multi-channel surround sound experiences defined by Dolby and DTS standards.

As with video, the audio decoder usually handles the actual decompression, while host software configures and controls the flow.

Other Broadcast Elementary Streams

Broadcast TV does not carry only video and audio. It can include several additional elementary stream components.

One important example is subtitles. DVB subtitles are defined as bitmap subtitles. These bitmaps are rendered and layered on top of the video content, synchronized with playback.

DVB also defines traditional teletext, which is still supported by many TV receivers.

ATSC defines its own closed captioning standard. Unlike DVB bitmap subtitles, ATSC captions are text-based. The receiver interprets the text description and decides how the final presentation should appear.

These additional elementary streams show that broadcast multiplexes are not only video and audio containers. They can carry many different media-related data types.

Conditional Access and Scrambling Standards

Conditional access systems and scrambling algorithms may be open or proprietary.

DVB Common Scrambling Algorithm defines a common scrambling algorithm. With such an approach, receivers can in principle be equipped to descramble content using the same algorithm, as long as they obtain the correct keys.

Proprietary systems such as NAGRA, Viaccess-Orca, Irdeto, Verimatrix and others take a different approach. They keep both the keys and the algorithms protected. This can provide a more secure system, but it requires specific software and hardware support inside the receiver.

Because of this, proprietary conditional access may require special set-top boxes or descrambling cards.

CI and CI+ standards provide a standardized way to connect conditional access modules, or CAMs, to TVs. These modules are inserted through a PCMCIA-style interface slot and enable proprietary descrambling operations inside otherwise standard TV devices.

Broadcast Receiver SoCs

Broadcast receivers often use dedicated system-on-chip platforms, although this is not a strict rule.

Broadcast-oriented SoCs usually include built-in support for MPEG-TS hardware demultiplexing, common audio and video formats, and descrambling functions. They also come with software SDKs that help unpack broadcast metadata and communicate with tuners and other processing blocks.

Common SoC vendors in this area include Broadcom, MediaTek and Amlogic.

The SoC is important because many broadcast operations are time-sensitive and data-intensive. Hardware assistance is often needed for reliable real-time reception, demuxing, decoding and presentation.

Satellite Broadcasting

When switching from terrestrial to satellite broadcasting, the main change is in the distribution part.

Satellite broadcasting standards are more globally aligned because satellites are not tied to local terrestrial tower networks in the same way. DVB-S and DVB-S2 are dominantly used.

Once the satellite signal is demodulated and available as a bitstream, the rest of the flow remains familiar. The stream is still typically MPEG-TS, using PES packets, common audio and video coding formats, PSI/SI metadata and conditional access mechanisms.

The main visible difference on the receiver side is that the device needs a satellite tuner.

Cable Broadcasting

Cable broadcasting also differs primarily at the distribution layer.

The signal characteristics inside a cable network are different from terrestrial or satellite transmission. Therefore, the types of errors and suitable modulation schemes also differ.

Cable infrastructure is installed and managed by cable companies, so each operator can choose the standard it wants to use. Most of the world uses DVB-C and DVB-C2. In North America, OpenCable standards from CableLabs are primarily used.

Again, after demodulation, the digital stream structure is usually the same familiar MPEG-TS-based structure.

Mapping Standards to the Signal Path

We can now follow the broadcast signal path and identify the formats and standards used at different stages.

In the production stage, the transport stream packaging step uses MPEG-TS. The output is a steady stream of TS packets, each 188 bytes long. When Reed-Solomon coding is added for error correction, the size typically grows to 204 bytes.

Each TS packet includes a header. The header starts with the synchronization byte 0x47, which allows the receiver to recognize TS packet boundaries. The header also includes a 13-bit PID, or packet identifier. The PID identifies which elementary stream or metadata table the TS packet payload belongs to.

At multiplexing, packets from different elementary streams are interleaved. Each TS packet carries payload associated with one specific elementary stream or table. The receiver later uses the PID values to separate them.

Timing and PCR

For uninterrupted decoding, all elementary streams must be delivered with enough data available before their presentation time.

MPEG-TS uses timing information to support synchronization. TS packets may carry a Program Clock Reference, or PCR, value in the adaptation field. PCR indicates the reference time used by the receiver to synchronize its clock.

Usually, one elementary stream is selected as the reference stream for timing. The PID of this stream is called the PCR PID. The receiver consults TS packets with this PID to synchronize its local clock with the timing of the incoming program.

PES Packets, DTS and PTS

Program stream packaging creates PES packets. Each elementary stream is packetized into PES, and each PES packet is then split into the necessary number of TS packets for transport.

PES packet sizes are variable because they depend on the type and amount of data they carry.

PES also contains key information for synchronization. In a real receiver, different processing blocks may require different amounts of time. Audio, video and subtitles may be decoded by different units, and their processing duration may not be exactly equal. Also, no two PES packets are delivered to the receiver at precisely the same instant.

The production stage solves this by packaging data early enough. PES packets are inserted into the MPEG-TS stream before the time at which their decoded results need to be presented.

This timing is defined relative to the PCR clock.

Each PES packet may carry two additional time markers:

- DTS, or Decoding Time Stamp, tells the receiver the PCR time by which the decoder must finish decoding the PES data.
- PTS, or Presentation Time Stamp, tells the receiver when the decoded result must be presented.

For example, after decoding a video PES packet, the video decoder may write the decoded frame into the GPU video layer at the correct presentation time.

The GPU composition delay, HDMI signalling delay and screen display delay exist, but they are usually predictable and small enough not to disturb the perceived output.

If the audio decoder respects PTS, the video decoder respects PTS, the subtitle decoder respects PTS and all refer to the same PCR clock, the user experiences synchronized media output. In practical terms, synchronization is acceptable if the delay is small enough to remain unnoticed, often considered to be within roughly 150 milliseconds.

Broadcast Metadata Structure

Broadcast metadata is carried inside MPEG-TS as PSI and SI tables.

In the stream, not all TS packets carry PES data. Some TS packets carry signalling tables. These tables are repeatedly broadcast in a data carousel fashion, meaning the same tables appear periodically in the stream.

PAT: Program Association Table

The most important starting point for metadata parsing is the PAT, or Program Association Table.

The PAT is carried in TS packets whose PID is always zero. It acts as the index table of the transport stream.

The central part of PAT is the program loop. This loop lists all services transported within the same MPEG-TS multiplex on the currently tuned frequency. For example, the multiplex may contain BBC and Eurosport.

For each service, the PAT provides a program number and the PID of the corresponding PMT table. The PAT does not contain all service details. Instead, it points the receiver to the PMTs, where detailed service information can be found.

The PAT also points to the NIT, or Network Information Table, by revealing its PID. The NIT contains information about the broader frequency range the receiver can use to access services that may be broadcast on other frequencies.

PMT: Program Map Table

The Program Map Table, or PMT, provides detailed information about one particular service.

PMTs are carried in TS packets with PIDs revealed by the PAT.

Each PMT references the PCR PID for the service, telling the receiver which elementary stream should be used as the timing reference.

The PMT also contains an elementary stream loop listing all elementary streams belonging to that service. For example, a BBC service may include one video elementary stream, one audio stream, another audio stream in a different language, and one subtitle stream.

Each elementary stream is packetized as PES and carried in TS packets identified by the PIDs listed in the PMT. Therefore, the audio PID identifies all TS packets carrying audio PES packets for that service. The video PID does the same for video.

The PMT also includes information about stream types and coding formats, which is required for decoder configuration.

Other PSI/SI Tables

Many additional tables exist and are also broadcast in carousel fashion.

The Service Description Table, or SDT, gives human-readable service information, such as service names.

The Conditional Access Table, or CAT, points to service keys and conditional access information needed for descrambling.

The Event Information Table, or EIT, provides schedule information for services. It is used to populate the electronic program guide in a DTV application.

The Application Information Table, or AIT, indicates whether a service has a corresponding interactive application or web portal, such as an HbbTV application.

Gathering broadcast metadata can therefore be complex. The receiver parses PAT to find PMTs. PMTs identify services and elementary streams. Service numbers are cross-referenced with SDT to obtain descriptions and with EIT to obtain scheduled events. Other tables provide conditional access, network and application information.

This means the receiver must parse and cross-reference many tables to obtain a complete picture of the transport stream contents.

PAT and PMT are standard MPEG tables. Many additional SI extensions depend on the broadcasting system, such as DVB. ATSC uses a different set of tables through PSIP for similar purposes.

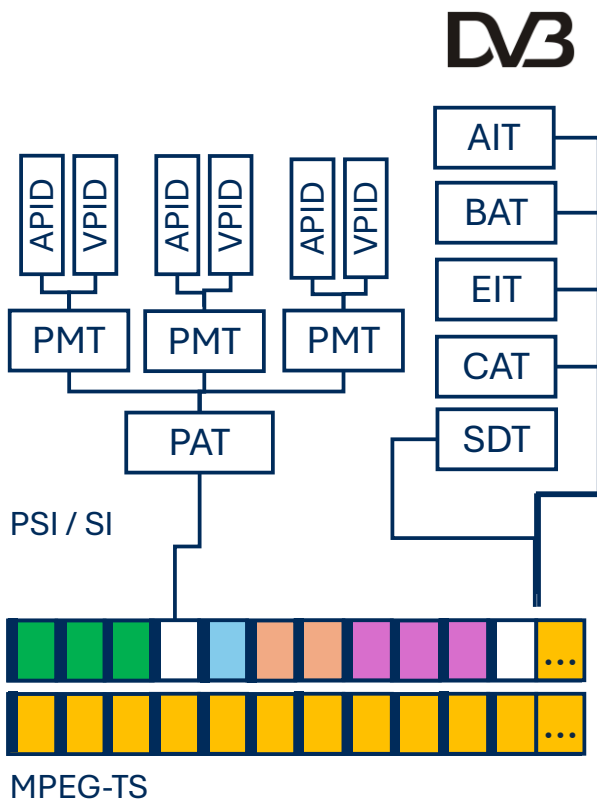


Figure 1. PSI/SI metadata within MPEG-TS

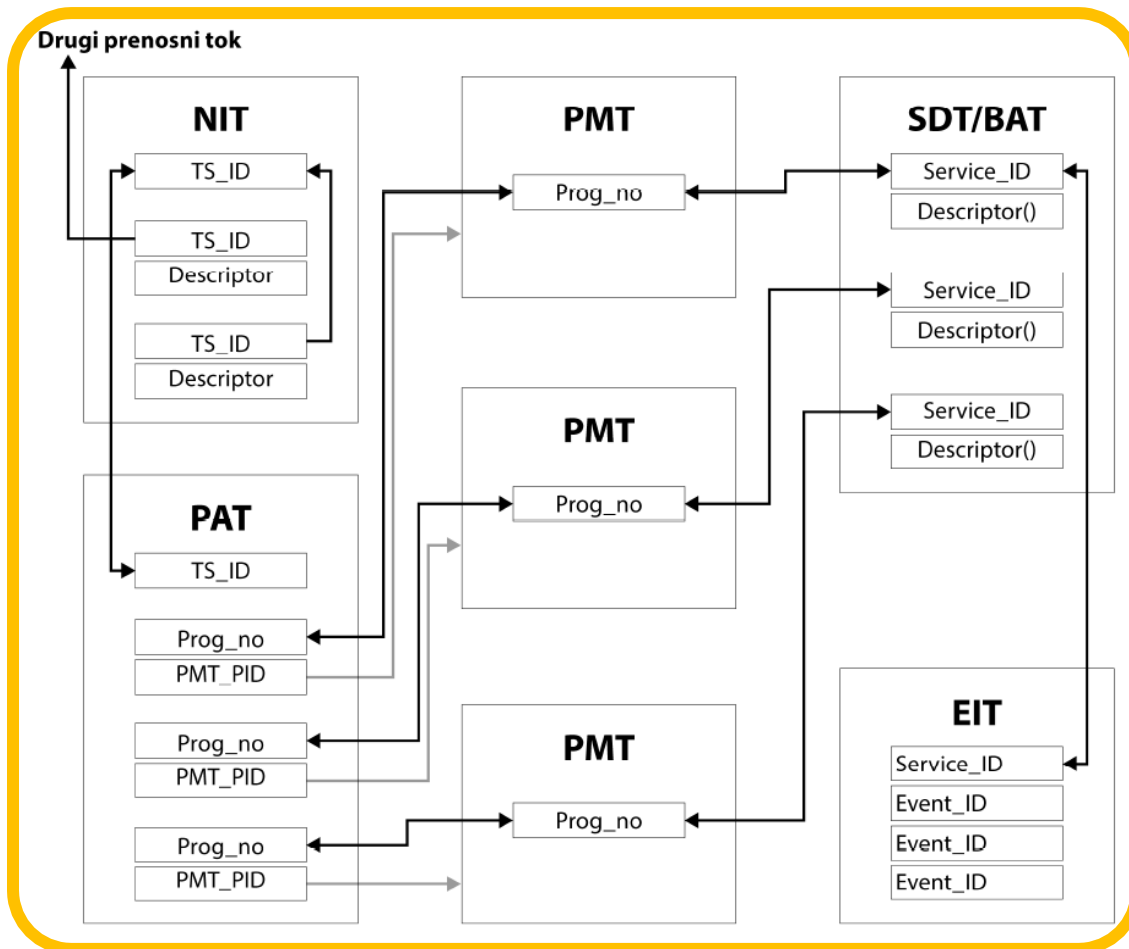


Figure 2. Cross-referencing between PSI and SI tables

Video Coding Inside the Container Hierarchy

We can now return to the program stream layer and look more closely at video coding.

As discussed earlier, video compression avoids transmitting every frame as a complete raw image. Instead, it transmits only the information needed to reconstruct visually relevant changes.

Video coding standards define a group of pictures, or GOP, as a decodable unit that can be transported in a PES packet.

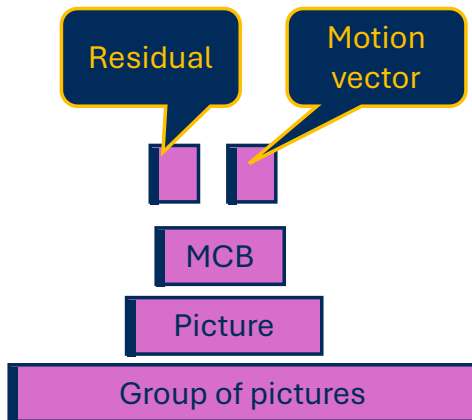
A GOP usually starts with a full picture called an I-frame. This frame is spatially coded and can be understood as conceptually similar to a JPEG image.

The I-frame is followed by P-frames and B-frames. These are predicted pictures described using macroblocks, motion vectors and residuals.

A P-frame can be reconstructed by referencing macroblock positions from a previous frame, applying motion vectors and adding decoded residual information.

This creates a deep encapsulation hierarchy. Residuals and motion vectors are packed into macroblocks. Macroblocks form predicted frames. Predicted frames and I-frames form a GOP. The GOP is packed into a PES packet. The PES packet is split into TS packets. Each TS packet carries a PID indicating that it belongs to the video elementary stream.

This layered structure is one of the central engineering ideas behind digital broadcast media.



A/V, e.g. MPEG-2, AVC, HEVC...



PES



MPEG-TS

Figure 3. Transport stream data hierarchy example

Scrambling and Conditional Access in Detail

Conditional access can be understood by looking at descrambling. Scrambling is essentially the inverse operation.

In a scrambled transport stream, the content payload is unreadable and unusable without authorization. TS packets carrying PES information are scrambled, but their headers remain readable. Signalling tables also remain accessible because the receiver needs metadata to discover how to obtain descrambling information.

A scrambled sequence can be descrambled if the receiver knows the control word. However, the control word is itself protected.

The receiver obtains an encrypted control word from stream metadata, for example from PSI sections. To decrypt the control word, it needs a service key. The service key is also found in the metadata, but it is encrypted as well.

To decrypt the service key, the receiver needs a user key. The user key may be embedded in a set-top box provided by a telecom operator, stored on a smart card inserted into the receiver, or derived from user credentials.

The user key is constructed so that it can decrypt only the service keys for services or packages to which the user is subscribed.

Once the service key is recovered, it can decrypt the control words for services in the package. Once the control word is known, the receiver can initialize the descrambler and descramble the transport stream.

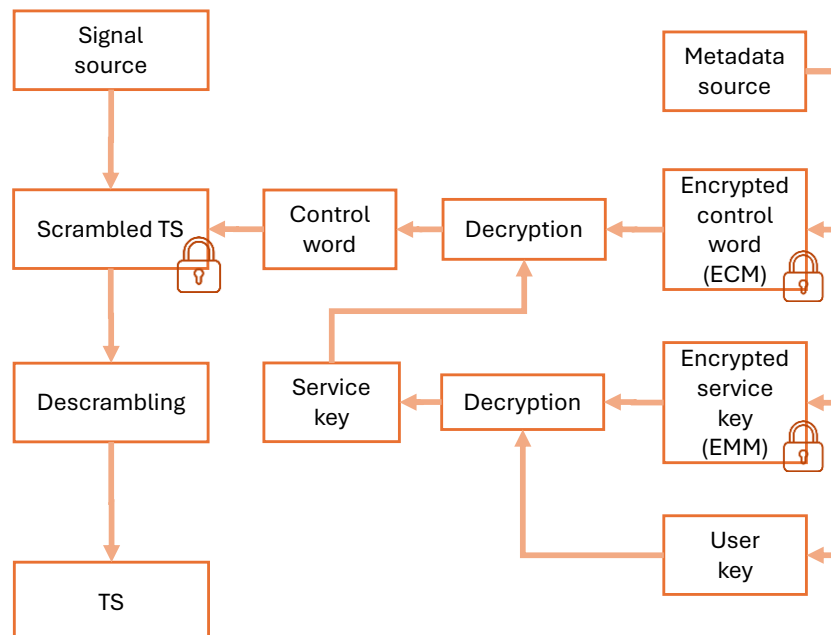


Figure 4. Descrambling with CAS and keys

ECM and EMM

In terms of specific data types, the encrypted control word is carried in an Entitlement Control Message, or ECM. The ECM is contained in a PSI section inside dedicated TS packets.

The encrypted service key is carried in an Entitlement Management Message, or EMM, also within PSI sections.

The receiver discovers ECM and EMM locations through metadata tables.

The process starts with the transport stream. TS packets with PID 0 carry the PAT. The PAT points to the PMT for a service. The PMT lists the elementary streams, such as audio and video, by their PIDs.

If the service is scrambled, the PMT also contains a CA descriptor. This descriptor points to the ECM, where the encrypted control word is found.

Separately, TS packets with PID 1 carry the CAT. The CAT is inserted by the provider of premium content packages, such as a telecom operator. It contains references to EMM tables from one or more conditional access systems.

If the user belongs to a supported conditional access system and has the correct subscription, the user key can decrypt the corresponding EMM. The EMM then provides the service key. The service key decrypts the ECM. The ECM provides the control word. The control word initializes the descrambler.

This chain explains how the receiver moves from user entitlement to actual content descrambling.

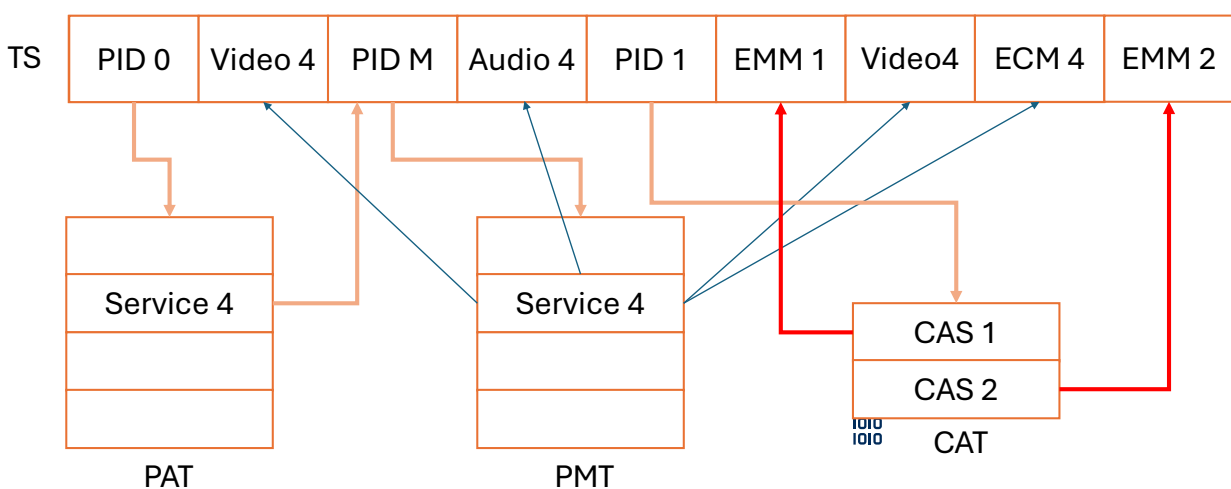


Figure 5. Keys within PSI metadata

Modulation and Signalling

The modulation and signalling layer is fully defined by the broadcasting standard. Different standards exist for terrestrial, satellite and cable systems, such as DVB-T/T2, DVB-S/S2, DVB-C/C2, ATSC, ISDB-T, DTMB and OpenCable.

These standards define how the digital bitstream is physically transmitted as a modulated signal and how the receiver should recover it.

Once demodulation is complete, the bitstream is interpreted through the common container and metadata structures already discussed.

Recap of Broadcast Standards and Formats

At the lowest transmission layer, broadcast systems use standards from groups such as DVB, ISDB, DTMB and ATSC. These standards define the physical transmission and modulation layer.

The transmitted modulated signal is converted into a bitstream. That bitstream is interpreted using container formats in two layers:

- MPEG-TS, where TS packets are the smallest digital transmission units in broadcast systems
- PES, where elementary streams such as video and audio are packetized into decodable units

This is enriched with metadata:

- PSI provides core information about programs, elementary streams and scrambling
- SI and PSIP provide additional service, scheduling and system information depending on the broadcasting standard

Elementary stream data is then coded using dedicated standards:

- Video codecs such as MPEG-2, H.264/AVC, H.265/HEVC and H.266/VVC
- Audio codecs such as AAC, HE-AAC, AC-3, Dolby and DTS formats
- Other formats such as DVB subtitles, teletext, closed captions and interactive application portal formats such as MHEG, BML and HbbTV

Together, these standards define the complete structure of broadcast digital media delivery.

Data	Video: MPEG-2, H.264 AVC, H.265 HEVC, H.266 VVC Audio: AAC, HE-AAC, AC3 Subtitles: DVB SU, ATSC CC Interactive: TTX, MHEG-5, BML, HbbTV
Metadata	MPEG-2 PSI: PAT, PMT, CAT DVB-SI: NIT, BAT, SDT, EIT, RST, TDT, TOT, SIT ATSC PSIP: MGT, VCT, RRT, STT, EIT, ETT ARIB SI: NIT, SDT, EIT, TOT, BIT, CDT, DIT, SIT
Containers	MPEG-TS PES
Transmission	Terrestrial: DVB-T, DVB-T2, ISDB-T, ATSC 1.0, ATSC 3.0, DTMB Satellite: DVB-S, DVB-S2, ISDB-S, ISDB-S3 Cable: DVB-C, DVB-C2, ISDB-C, OpenCable / CableLabs

Figure 6. Most common standards and formats in broadcasting

Applying These Details to Project Work

This lecture adds more detail to the first project, especially for use cases involving broadcast TV signals.

When revisiting a dataflow diagram, each arrow should now be inspected more carefully. The question is no longer only “which block sends data to which block?” but also “which standard, format and data type is used on this connection?”

For example, a connection may carry MPEG-TS packets, PES packets, PSI sections, decoded PCM audio, decoded video frames, subtitle bitmaps, ECMs, EMMs, or HDMI output.

Students should add callouts to their diagrams specifying the relevant standard, format or data type used at each step.

The same applies to UML sequence diagrams. Now that specific standards and data types are known, command and data exchanges can be made more precise. A command may configure a tuner for a DVB-T2 frequency, instruct a demux to filter a specific PID, configure a decoder based on a stream type from the PMT, or initialize descrambling using information discovered through CAT, ECM and EMM processing.

The goal is to move the architecture diagrams one step closer to the actual digital media world.

Closing Perspective

This lecture connected the conceptual broadcast pipeline to concrete standards and data formats.

We saw that physical transmission is governed by broadcast standards such as DVB, ATSC, ISDB and DTMB. Once the signal becomes a digital bitstream, MPEG-TS and PES define the transport structure. PSI, SI and PSIP provide the metadata needed to understand the multiplex. Audio, video, subtitles and other elementary streams use their own coding and representation standards. Conditional access adds another layer of metadata and cryptographic control using ECM, EMM, service keys and control words.

This combination of standards is what makes broadcast TV interoperable, scalable and technically manageable.

In the next lecture, the same type of analysis will be applied to broadband distribution, where Internet protocols, streaming formats and OTT-specific technologies define a different but equally important media delivery architecture.